

Integration of Vector functions

1. If $\vec{A}(t) = (3t^2 - 2t)\hat{i} + (6t - 4)\hat{j} + (4t)\hat{k}$
evaluate $\int_2^3 \vec{A}(t) dt$

$$\Rightarrow \hat{i} \int_2^3 (3t^2 - 2t) dt + \hat{j} \int_2^3 (6t - 4) dt + \hat{k} \int_2^3 4t dt$$

$$\Rightarrow \hat{i} \left[\frac{3t^3}{3} - \frac{2t^2}{2} \right]_2^3 + \hat{j} \left[\frac{6t^2}{2} - 4t \right]_2^3 + \hat{k} \left[\frac{4t^2}{2} \right]_2^3$$

$$\Rightarrow \hat{i} [27 - 9 - 8 + 4] + \hat{j} [18 - 12 - 12 + 8] + \hat{k} [18 - 8]$$

$$\Rightarrow 14\hat{i} + 11\hat{j} + 10\hat{k}$$

2. The acceleration of a particle at time t is given by
 $\vec{a} = 18\cos 3t \hat{i} - 8\sin 2t \hat{j} + 6t \hat{k}$. If the velocity $\vec{v}$
and displacement $\vec{r}$ be zero at $t=0$. Find $\vec{v}$ and
 $\vec{r}$ at any point t .

Line integrals.

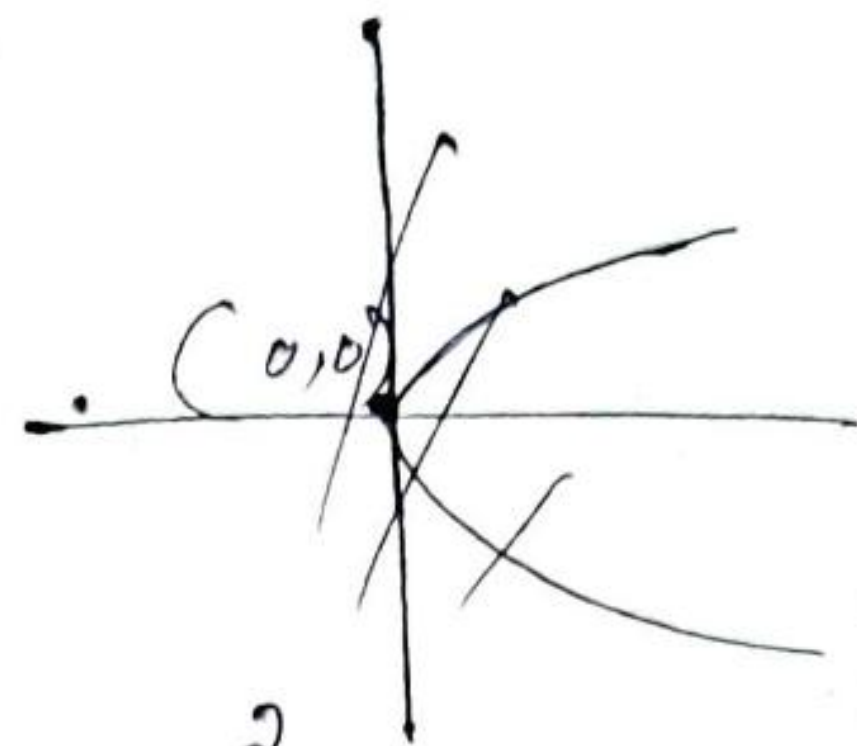
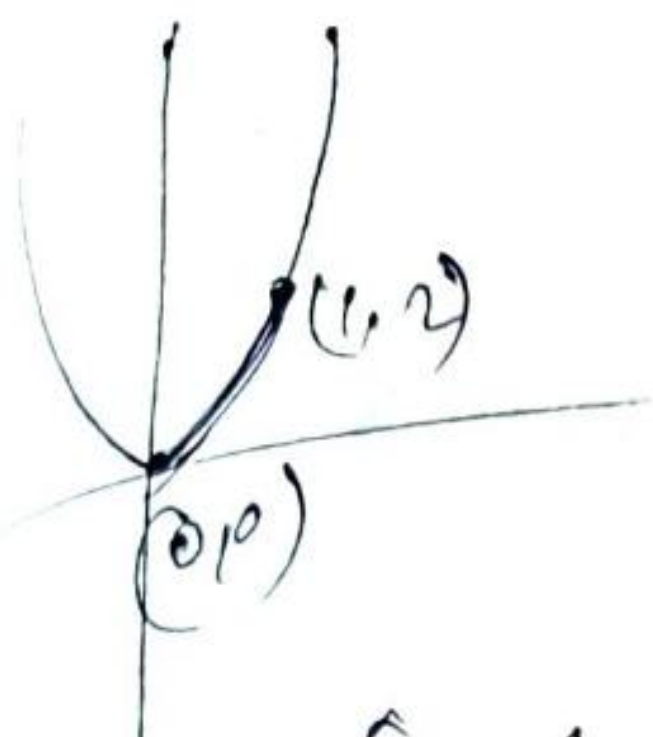
Any integral which is to be evaluated along a curve is called line integral and is denoted by $\int_C \vec{F} d\vec{r}$ or $\int_C \vec{F} \cdot \frac{d\vec{r}}{dt} dt$

Work done by a force

Let $\vec{F}$ represent the force acting on a particle moving along an arc AB. The work done during a small displacement $d\vec{R}$ is $\int \vec{F} \cdot d\vec{R}$

The total work done by $\vec{F}$ during displacement from A to B is given by $\int_A^B \vec{F} \cdot d\vec{R}$

$\vec{F} = 3xy\hat{i} - y^2\hat{j}$ evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the arc of the parabola $y = 2x^2$ from $(0,0)$ to $(1,2)$



$$\int_C 3xy\hat{i} + (-y^2)\hat{j} = \int_0^1 3xy dx + \int_0^2 -y^2 dy$$

$$\Rightarrow \int_0^1 3x \times 2x^2 dx + \int_0^2 -y^2 dy = \int_0^1 6x^3 dx - \int_0^2 y^2 dy$$

$$\Rightarrow \left[\frac{6x^4}{4} \right]_0^1 - \left[\frac{y^3}{3} \right]_0^2 = \frac{6}{4} - \frac{8}{3} = \frac{18-32}{12}$$

$$y = 2x^2$$

$$dy = 4x dx$$

$$\int_C 3xy\hat{i} + (-y^2)\hat{j} = \int_0^1 3x \times (2x)^2 dx - \int_0^1 (2x^2)^2 \times 4x dx$$

$$\Rightarrow \int_0^1 6x^3 dx - \int_0^1 16x^5 dx$$

$$\Rightarrow \left[\frac{6x^4}{4} \right]_0^1 - \left[\frac{16x^6}{6} \right]_0^1 = \frac{6}{4} - \frac{16}{6} = \frac{9-16}{6} = -\frac{7}{6}$$

2. A vector field is given by $\vec{F} = \sin y \hat{i} + x(1+\cos y)\hat{j}$. Evaluate the line integral over the circular path given by $x^2 + y^2 = a^2$, $z=0$.

$$M = \sin y$$

$$N = x(1+\cos y)$$

$$\frac{\partial M}{\partial y} = \cos y$$

$$\frac{\partial N}{\partial x} = (1+\cos y)$$

$$\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = 1 + \cos y - \cos y = 1$$

$$\int \int_D 1 dx dy$$

$$x = a \cos \theta$$

$$y = a \sin \theta$$

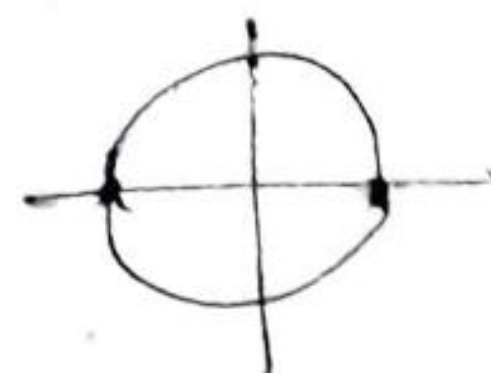
$$\int_0^{2\pi} \int_0^a 1 \cdot r dr d\theta$$

$$= \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^a d\theta$$

$$= \int_0^{2\pi} \frac{a^2}{2} d\theta$$

$$= \frac{a^2}{2} \int_0^{2\pi} 1 d\theta$$

$$= \frac{a^2}{2} \cdot 2\pi = \pi a^2$$



$$\int_a^b \sqrt{a^2 - x^2} dx \quad \text{let } x = a \sin \theta$$

$$\frac{dx}{d\theta} = a \cos \theta$$

$$\int_a^b \sqrt{a^2 - a^2 \sin^2 \theta} \times a \cos \theta d\theta$$

$$\int_a^b \sqrt{a^2 - x^2} dx \quad \text{let } x = a \cos \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$\int_a^b \sqrt{a^2 - a^2 \cos^2 \theta} \times -a \sin \theta d\theta$$

$$1) \int_0^{\pi/2} a \sin \theta \times -a \sin \theta d\theta \Rightarrow 4a^2 \int_0^{\pi/2} -\sin^2 \theta d\theta$$

$$2) \int_0^{\pi/2} 4a^2 \times 1 \times \pi/2 = a^2 \pi$$

3. Find the work done by the force $\vec{F} = x\hat{i} - z\hat{j} + 2y\hat{k}$ in the displacement along the closed path C consisting of the segments C_1, C_2, C_3 where.

On C_1 , $0 \leq x \leq 1$, $y = x$, $z = 0$

On C_2 , $0 \leq z \leq 1$, $x = 1$, $y = 1$

On C_3 , $1 \geq x \geq 0$, $y = z = x$

Work done = $\int_C x dx - z dy + 2y dz$

$$= \int_{C_1} x dx - z dy + 2y dz + \int_{C_2} x dx - z dy + 2y dz + \int_{C_3} x dx - z dy + 2y dz$$

$$\Rightarrow \int_0^1 x dx - z dy + 2y dz + \int_0^1 1 dx - z dy + 2y dz + \int_1^0 x dx - x dx + 2x dx$$

$$\int x dx + \int 2 dz + \int 2 x dx \Rightarrow \left[\frac{x^2}{2} \right]_0^1 + [2z]_0^1 + \left[\frac{2}{2} \right]_0^1$$

$$\Rightarrow \frac{1}{2} + 2 + 1 \Rightarrow \frac{7}{2}$$

Surface integral

Any integral which is to be evaluated over a surface is called a surface integral (denoted by $\iint_S \vec{F} \cdot \hat{n} ds$) ($\hat{n}$ outward drawn normal). In order to evaluate surface integrals it is convenient to express them as double integrals taken over the orthogonal projection of S on one of the co-ordinate planes.

Let R be the orthogonal projection of S on the xy -plane. Let $\hat{n} = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$ where $\cos \alpha$, $\cos \beta$, $\cos \gamma$ are the direction co-sine of $\hat{n}$.
 $dxdy =$ projection of ds on the xy plane

$$ds = \frac{dxdy}{\cos \gamma} = \frac{dxdy}{|\hat{k} \cdot \hat{n}|} \quad \cos \gamma = |\hat{k} \cdot \hat{n}|$$

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_R \vec{F} \cdot \hat{n} \frac{dxdy}{|\hat{k} \cdot \hat{n}|}$$

Volume integrals

Any integral which is to be evaluated over a volume is called a volume integral.

If V is a volume bounded by a surface S , then the triple integrals $\iiint_V \phi dv$ and $\iiint_V \vec{F} dv$ are called volume integrals.

$$\iiint_V \phi dv = \iiint_V \phi(x, y, z) dxdydz$$

If $\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$ then

$$\iiint_V \vec{F} dv = \hat{i} \iiint_V F_1(x, y, z) dxdydz + \hat{j} \iiint_V F_2(x, y, z) dxdydz + \hat{k} \iiint_V F_3(x, y, z) dxdydz$$

Gauss Divergence Theorem (Relation betⁿ surface and vol integrals)

If $\vec{F}$ is a vector point function having continuous first order partial derivatives in the region V bounded by a closed surface S , then.

$$\iiint_V \nabla \cdot \vec{F} dV = \iint_S \vec{F} \cdot \hat{n} dS, \text{ where } \hat{n} \text{ is the outwards drawn unit normal vector to the surface } S.$$

Stokes' Theorem (Relation betⁿ line and surface integrals)

If S be an open surface bounded by a closed curve C and $\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$ be any vector point function having continuous first order partial derivatives, then

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} dS \text{ where } \hat{n} \text{ is a unit normal vector at any point of } S \text{ drawn in the sense in which a right handed screw would advance when rotated in the sense of description of } C.$$

1. Evaluate $\iint_S \vec{A} \cdot \hat{n} dS$ where $\vec{A} = (x+y^2)\hat{i} - 2xy\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x+y+2z=6$ in the first octant

$$\nabla \cdot (2x+y+2z) = 2\hat{i} + \hat{j} + 2\hat{k}$$

$$\hat{n} = \frac{2\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{4+1+4}} = \frac{2\hat{i} + \hat{j} + 2\hat{k}}{3}$$

$$dS = \frac{dxdy}{\hat{n} \cdot \hat{k}}$$

$$\vec{A} \cdot \hat{n} = \left[(x+y^2)\hat{i} - 2xy\hat{j} + 2yz\hat{k} \right] \cdot \left(\frac{2\hat{i} + \hat{j} + 2\hat{k}}{3} \right)$$

$$= \frac{1}{3} [2(x+y^2) - 2x + 2 \times 2yz]$$

$$\Rightarrow \frac{1}{3} (2x + 2y^2 - 2x + 4yz)$$

$$\Rightarrow \frac{1}{3} (2y^2 - 2xz) \Rightarrow \frac{2y^2}{3} + \frac{4yz}{3}$$

$$\Rightarrow \frac{2y^2}{3} + \frac{4}{3} \left(\frac{6-2x-y}{2} \right) y$$

$$\frac{2y^2}{3} + \frac{2y}{3}(6-2x-y) \Rightarrow \frac{2}{3} [y^2 + 6y - 2xy - y^2]$$

$$\Rightarrow \frac{2}{3} [6y - 2xy] = \frac{4y}{3} (3-x)$$

$$\int_0^3 \int_0^{6-2x} \left(\frac{4y}{3} - \frac{4xy}{3} \right) \frac{dxdy}{2/3}$$

$$2x+y=6$$

$$\begin{cases} x=3 \\ x=0 \end{cases}$$

$$|\hat{n} \cdot \hat{n}| = \frac{2}{3}$$

$$\int_0^3 \int_0^{6-2x} 4y - \frac{4xy}{3} dx$$

$$\int_0^3 \int_0^{6-2x} 2 \frac{4y^2}{2} - \frac{4}{3} x \frac{y^2}{2} dy$$

$$\Rightarrow \frac{3}{2} \int_0^3 2x(6-2x)^2 dx - \frac{2}{3} x(6-2x)^2 dx$$

$$\Rightarrow \frac{3}{2} \int_0^3 2(6-2x)^2 dx - \frac{2}{2} \int_0^3 x(6-2x)^2 dx$$

$$\int_0^3 \int_0^{6-2x} \frac{4y}{3} (3-x) \frac{dxdy}{2/3} = \frac{3}{2} \int_0^3 dx \int_0^{6-2x} (3-x) \frac{4y}{3} dy$$

$$\Rightarrow \frac{4}{3} \int_0^3 dx \times (3-x) \frac{y^2}{2} \Rightarrow \frac{2}{2} \int_0^3 (6-2x)^2 (3-x) dx$$

$$\Rightarrow \int_0^3 (6-2x)^2 dx \Rightarrow \int_0^3 (6-x)^3 dx$$

$$\Rightarrow \left[\frac{(3-x)^4}{-1 \times 1} \right]_0^3 = \left[\frac{(3-x)^3}{-1} \right]_0^3 \Rightarrow 81$$

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2. Evaluate $\iint \vec{A} \cdot \hat{n} \, ds$ where $\vec{A} = x\hat{i} + y\hat{j} + z\hat{k}$ is the surface of the cylinder $x^2 + y^2 = 16$ included in the octant between $z=0$ and $z=5$.

$\vec{A}$ = Surface $x^2 + y^2 = 16$

vector form $= 2x\hat{i} + 2y\hat{j}$ ($\nabla \cdot S$)

$$\hat{n} = \frac{(2x)^2 + (2y)^2}{\sqrt{4x^2 + 4y^2}} = \frac{2x\hat{i} + 2y\hat{j}}{2 \times 4} = \frac{x\hat{i} + y\hat{j}}{4}$$

$$ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = \frac{dx dy}{1/4}$$

$$\vec{A} \cdot \hat{n} = (z\hat{i} + x\hat{j} - 3y^2z\hat{k}) \cdot \left(\frac{x\hat{i}}{4} + \frac{y\hat{j}}{4} \right)$$

$$= \frac{xz}{4} + \frac{xy}{4} \Rightarrow \frac{x}{4}(y+z)$$

$$\iint \vec{A} \cdot \hat{n} \, ds \Rightarrow \iint \frac{x}{4}(y+z) \times \frac{dx dy}{1/4}$$

$$\Rightarrow \int_{z=0}^5 \int_{y=0}^4 (y+z) \, dy \, dz$$

$$\Rightarrow \int_0^5 dz \int_0^4 (y+z) \, dy \Rightarrow \int_0^5 dz \left[\frac{y^2}{2} + zy \right]_0^4$$

$$\Rightarrow \int_0^5 dz \left(\frac{16}{2} + 4z \right) \Rightarrow \int_0^5 (8 + 4z) \, dz$$

$$\Rightarrow \left[8z + \frac{4z^2}{2} \right]_0^5 \Rightarrow 8 \times 5 + 2(5)^2$$

$$\Rightarrow 40 + 50 = 90$$

3. If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$ then evaluate

$\iiint_V \nabla \cdot \vec{F} \, dv$ where V is bounded by the plane $x=0, y=0,$

$z=0$ and $2x + 2y + z = 4$

$$\nabla \cdot \vec{F} = 4x - 2x \Rightarrow 2x$$

$$\int_0^2 \int_0^{4-2y-2x} 2x \, dz \, dy \, dx$$

$$\int_0^2 \int_0^{4-2y-2x} 2x \, dz \, dy \Rightarrow \int_0^2 \int_0^{2-x} dy \times [2xz]_0^{4-2y-2x}$$

$$\int_0^2 2x(4-2y-2x) \, dy \Rightarrow \int_0^2 \int_0^{2-x} (8x - 4xy - 4x^2) \, dy$$

$$\int_0^2 [8xy]_0^{2-x} - \left[\frac{4xy^2}{2} \right]_0^{2-x} - [4x^2y]_0^{2-x}$$

$$(2-x) - 2x(2-x)^2 - 4x^2(2-x) \, dx$$

$$-8x^2 - 2x(4+x^2-4x) - 8x^2 + 4x^3 \, dx$$

$$8x^2 - 8x - 2x^3 + 8x^2 - 8x^2 + 4x^3 \, dx$$

$$(8x^2 + 8x) \, dx \Rightarrow \left[\frac{8x^3}{3} \right]_0^2 - \left[\frac{8x^3}{3} \right]_0^2$$

$$+ \left[\frac{8x^2}{2} \right]_0^2$$

$$= \frac{8}{3} \times 2^3 + 4 \times 2^2 = 8 - 21.33 + 16$$

$$= 2.67$$

$\hat{i} = x\hat{j} + y^2\hat{k}$ evaluate $\iiint \vec{r} \cdot d\vec{v}$, where bounded by the surface!

$$x=2, y \geq 0, z = x^2, z=4$$

$$\int_{x^2}^4 \int_0^4 \int_0^{2.64} x \, dx \, dy \, dz$$

$$\int_0^2 \int_0^4 \int_{x^2}^4 y^2 \, dz \, dy \, dx$$

2. Evaluate $\iint_S \vec{A} \cdot \hat{n} \, dS$ where
and S is the surface of the
included in the first Octant

$\vec{A} = z\hat{i} + x\hat{j} - 3y^2\hat{k}$
Cylinder $x^2 + y^2 = 4$
between $z=0$ and $z=4$.

$$S: \nabla(x^2 + y^2) = 2x\hat{i} + 2y\hat{j}$$

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{(2x)^2 + (2y)^2}} = \frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\frac{2x\hat{i} + 2y\hat{j}}{2\sqrt{x^2 + y^2}} = \frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\hat{n} \cdot \hat{i} = \frac{x}{2}$$

$$\int_0^2 \int_0^6 \int_{\sqrt{4-y^2}}^4 2xz \, dz \, dy \, dx = \int_0^2 \int_0^6 \int_{\sqrt{4-y^2}}^4 2xz \, dz$$

$$\Rightarrow \int_0^2 \int_0^6 \left[\frac{2xz^2}{2} \right]_{\sqrt{4-y^2}}^4 dy \, dx \Rightarrow \int_0^2 \int_0^6 x(4^2 - x^2) dy \, dx$$

$$\Rightarrow \int_0^2 \int_0^6 (16x - x^3) dy \, dx$$

$$\Rightarrow \int_0^2 [16xy - x^3y]_0^6 dx \Rightarrow \int_0^2 (16x \cdot 6 - x^3 \cdot 6) dx$$

$$\Rightarrow 6 \int_0^2 (16x - x^3) dx \Rightarrow 6 \left[\frac{16x^2}{2} - \frac{x^4}{4} \right]_0^2$$

$$\Rightarrow 6 \left[\frac{16 \times 4}{2} - \frac{16}{4} \right] \Rightarrow 6(32 - 4)$$

$$\Rightarrow 6 \left[\frac{16 \times 2^2}{2} - \frac{2^4}{4} \right] = 6[32 - 4]$$

$$= 128.04$$

$$\int_0^2 \int_0^6 \int_{x^2}^4 x \, dx \, dy \, dz \Rightarrow -j \int_0^2 dx \int_0^6 dy \int_{x^2}^4 x \, dz$$

$$\Rightarrow -j \int_0^2 dx \int_0^6 dy \times [xz]_{x^2}^4 \Rightarrow -j \int_0^2 dx \int_0^6 dy \times 2[4 - x^2]$$

$$\Rightarrow -j \int_0^2 dx \int_0^6 (4x^2 - x^3) \, dy \Rightarrow -j \int_0^2 dx \times 4x \times 6$$

$$\Rightarrow -j \int_0^2 dx \int_0^6 dy \times 4x - x^3 \Rightarrow -j \times 6 \int_0^2 4x - x^3 \, dx$$

$$= -j \times 6 \left[\frac{4x^2}{2} - \frac{x^4}{4} \right]_0^2 = -6j \left[2 \times 4 - \frac{16}{4} \right]$$

$$\Rightarrow -6j \times 4 = -24j$$

$$+k \int_0^2 \int_0^6 \int_{x^2}^4 y^2 \, dx \, dy \, dz \Rightarrow k \int_0^2 dx \int_0^6 dy \int_{x^2}^4 y^2 \, dz$$

$$\Rightarrow k \int_0^2 dx \int_0^6 dy \times [y^2 z]_{x^2}^4 \Rightarrow k \int_0^2 dx \int_0^6 dy \times y^2(4 - x^2)$$

$$\Rightarrow k \int_0^2 dx \int_0^6 (4y^2 - x^2 y^2) \, dy \Rightarrow k \int_0^2 dx \times \left[\frac{4y^3}{3} - x^2 \frac{y^3}{3} \right]_0^6$$

$$\Rightarrow k \frac{6^3}{3} \int_0^2 (4 - x^2) \, dx \Rightarrow 72k \int_0^2 4 - x^2 \, dx$$

$$\Rightarrow 72k \times \left[4x - \frac{x^3}{3} \right]_0^2 \Rightarrow 72k [8 - \frac{8}{3}] = 384k$$

Ans $128.04\hat{i} - 24\hat{j} + 384.48\hat{k}$

5. Verify divergence theorem for $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ taken over the rectangular parallelepiped taken $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$.

Divergence theorem:
$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dV$$

$$\nabla \cdot \vec{F} = 2x + 2y + 2z \Rightarrow 2(x+y+z)$$

$$\iiint_V \nabla \cdot \vec{F} \, dV \Rightarrow \int_0^c \int_0^b \int_0^a 2(x+y+z) \, dx \, dy \, dz$$

$$2 \int_0^c \int_0^b \int_0^a x \, dx \, dy \, dz \Rightarrow 2 \int_0^c \int_0^b \left[\frac{x^2}{2} \right]_0^a \, dy \, dz$$

$$\Rightarrow 2 \times \frac{a^2}{2} \times 2 \int_0^c \int_0^b (x+y+z) \, dy \, dz$$

$$2 \int_0^c \int_0^b \left(\frac{a^2}{2} + ya + za \right) \, dy \, dz \Rightarrow 2 \int_0^c \int_0^b \left(\frac{a^2}{2} + ya + za \right) \, dy \, dz$$

$$\Rightarrow 2 \int_0^c \left(\frac{a^2}{2} y + \frac{y^2}{2} a + zya \right) \Big|_0^b \, dz$$

$$\Rightarrow 2 \int_0^c \left(\frac{a^2 b}{2} + \frac{b^2 a}{2} + zab \right) \, dz$$

$$\Rightarrow 2 \int_0^c \left[\frac{a^2 b}{2} z + \frac{b^2 a}{2} z + \frac{z^2}{2} ab \right] \Big|_0^c \, dz$$

$$\Rightarrow \frac{2}{2} [a^2 bc + b^2 ac + abc^3] \Rightarrow abc [a + b + c]$$

$$S_1 = ABCD$$

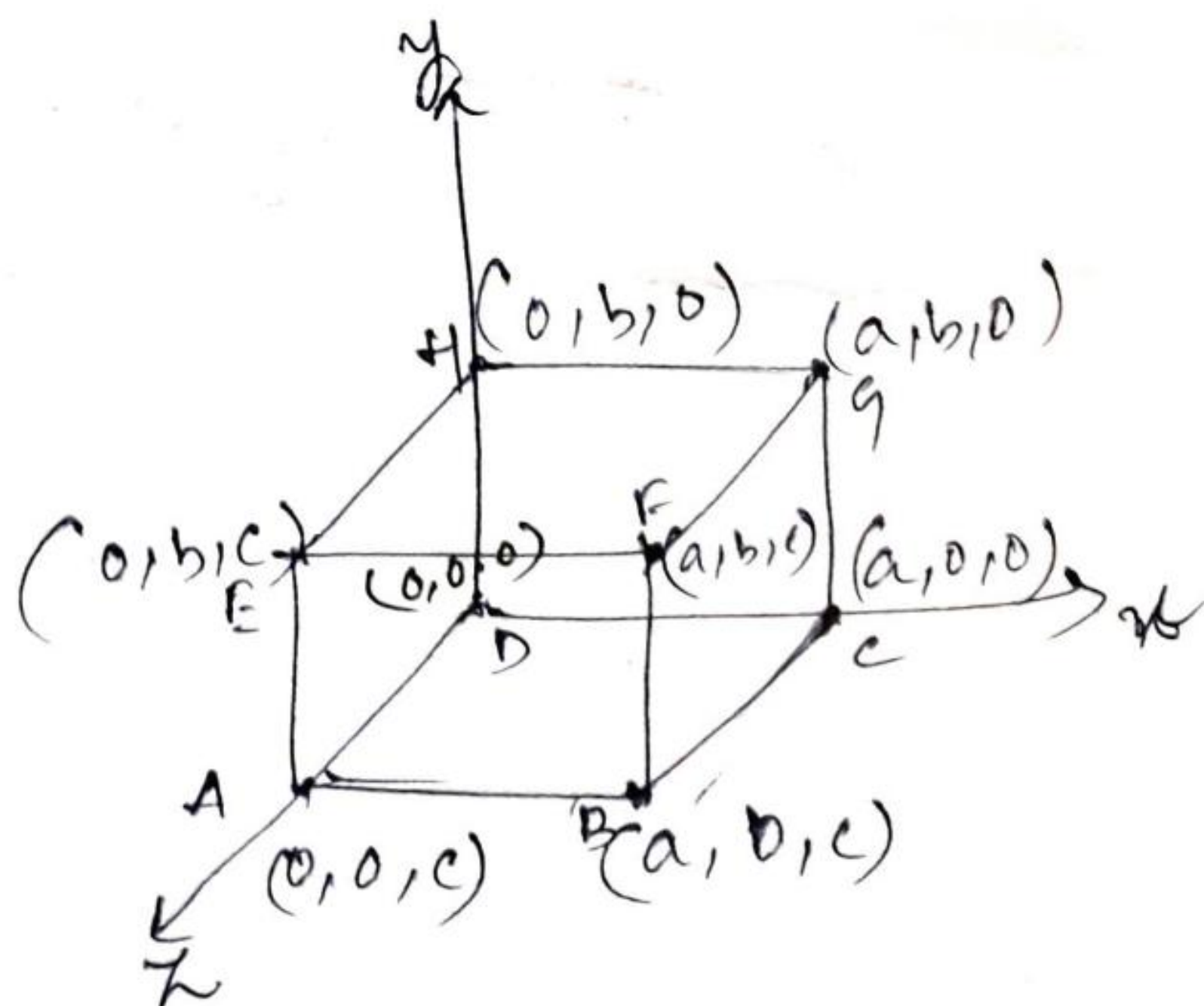
$$S_2 = HAGE$$

$$S_3 = EABF$$

$$S_4 = HGFCD$$

$$S_5 = HFEAD$$

$$S_6 = FGCB$$



$$\cancel{ds = dxdydz}$$

$$ds = \frac{dxdz}{|\hat{n} \cdot \hat{j}|}$$

$$dm = \int dz \cdot \rho \left(\frac{za^2}{2} \right)_0^a$$

$$\frac{1}{2} \left(\frac{x^2}{2} \right)_0^c = \frac{a^2 c^2}{4}$$

$$y = b$$

$$y^2 - za = b^2 - za$$

$$) dxdz \Rightarrow \int_0^c dz \int_0^b (b^2 - za) dx$$

$$\left(x - \frac{za^2}{2} \right)_0^a \Rightarrow \int_0^c dz \left[b^2 a - \frac{za^2}{2} \right]$$

$$\left(\frac{z^2 a^2}{4} \right)_0^c, \quad b^2 ac - \frac{a^3 c^2}{4}$$

BF-

$$= c \cdot \hat{n} = \hat{k}$$

$$= (z^2 - xy) \Rightarrow c^2 - xy$$

$$(c^2 - xy) dxdy \Rightarrow \int_0^b dy \int_0^a (c^2 - xy) dx$$

$$xy \left(c^2 x - \frac{x^2}{2} y \right)_0^a \Rightarrow \int_0^b \left(ac^2 - \frac{ya^2}{2} \right) dy$$

$$ac^2 b - \frac{b^3 a^2}{6}$$

$$S_4 = \cancel{HADC} \quad S_5 = H9CD$$

$$z=0 \quad \hat{n} = -\hat{k}$$

$$-\int_0^b \int_0^a (x^2 - xy) dy dx \Rightarrow \int_0^b \int_0^a xy dy dx \quad \cdot \quad \int_0^b dy \int_0^a xy dx$$

$$= \int_0^b dy \times \left[\frac{yx^2}{2} \right]_0^a \Rightarrow \int_0^b \frac{a^2}{2} y dy \Rightarrow \frac{a^2}{2} \times \frac{b^2}{2} = \frac{a^2 b^2}{4}$$

$$S_5 = H9AD$$

$$c \quad b \quad \hat{n} = -\hat{i} \quad x=0$$

$$\int_0^c \int_0^b yz - x^2 dy dz \Rightarrow \int_0^c dz \int_0^b yz dy$$

$$\Rightarrow \int_0^c dz \times \left[\frac{y^2 z}{2} \right]_0^b \Rightarrow \int_0^c \frac{b^2}{2} z dz \Rightarrow \frac{b^2}{2} \times \frac{c^2}{2} = \frac{b^2 c^2}{4}$$

$$S_6 = F9CB$$

$$\hat{n} = \hat{i} \quad x=a$$

$$\int_0^c \int_0^b (x^2 - yz) dx dy \Rightarrow \int_0^c \int_0^b (a^2 - yz) dy dz$$

$$\Rightarrow \int_0^c dz \int_0^b (a^2 - yz) dy \Rightarrow \int_0^c dz \times \left(a^2 y - \frac{y^2 z}{2} \right)_0^b$$

$$\Rightarrow \int_0^c \left(a^2 b - \frac{b^2 z}{2} \right) dz \Rightarrow \left[a^2 bz - \frac{b^2 z^2}{4} \right]_0^c$$

$$\Rightarrow a^2 bc - \frac{b^2 c^2}{4}$$

$$S_1 + S_2 + S_3 + S_4 + S_5 + S_6$$

$$\frac{a^2 c^2}{4} + b^2 ac - \frac{a^2 c^2}{4} + ac^2 b - \frac{a^2 b^2}{4} + \frac{a^2 b^2}{4} + \frac{b^2 c^2}{4}$$

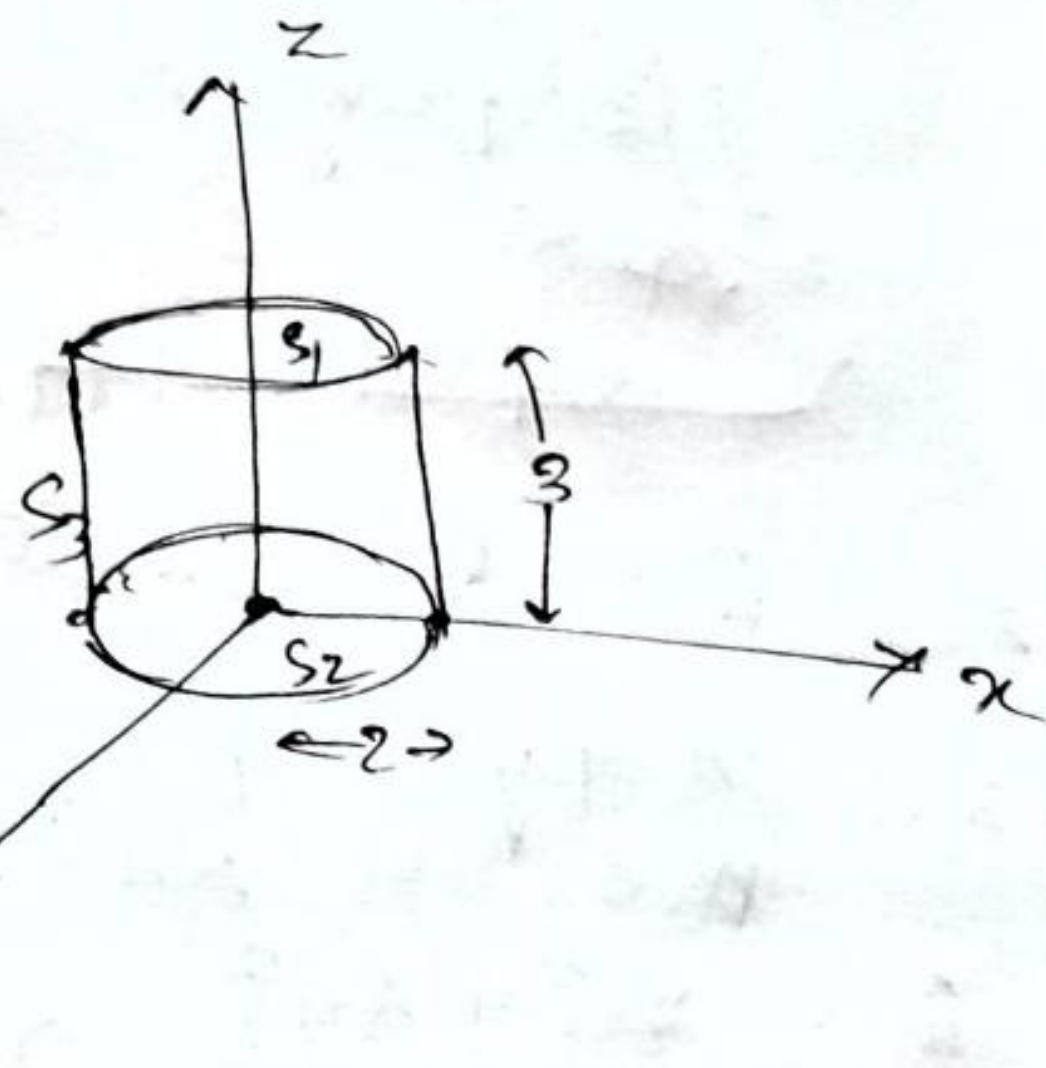
$$+ a^2 bc - \frac{b^2 c^2}{4} \Rightarrow abc(a+b+c)$$

6. Verify divergence theorem for $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by the cylinder $x^2 + y^2 = 4$, $x=0$, $z=3$.

$$\iiint_V \nabla \cdot \vec{F} \, dv = \iint_S \vec{F} \cdot \hat{n} \, ds$$

$$\nabla \cdot \vec{F} = 4 - 2y + 2z$$

$$\iiint_V \nabla \cdot \vec{F} \, dv = \int_0^3 \int_0^{2\pi} \int_0^2 (4 - 2y + 2z) \, r \, dr \, d\theta \, dz$$



$$\int_0^3 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^2 (4 - 2y + 2z) \, dy \, dx \, dz$$

$$\Rightarrow \int_0^3 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^2 (4 - 2y + 2z) \, dz \, dy \, dx$$

$$\Rightarrow \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^2 dy \times \left[4z - 2yz + \frac{2z^2}{2} \right]_0^3$$

$$\Rightarrow \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dy \times [12 - 6y + 9] = \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 6y) \, dy$$

$$\Rightarrow \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \times \left[21y - \frac{6y^2}{2} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$$

$$\Rightarrow \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \times 21 \times 2\sqrt{4-x^2} - 3(4-x^2 + 4-x^2)$$

$$\Rightarrow \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 42\sqrt{4-x^2} \, dx - 3 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 8 - 2x^2 \, dx$$

$$x = 2 \sin \theta$$

$$dx = 2 \cos \theta \, d\theta$$

$$\Rightarrow 2 \times 42 \times 2 \times \int_0^{\pi/2} \cos^2 \theta \, d\theta - 3 \left[8x - \frac{2x^3}{3} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$$

$$\Rightarrow \frac{2 \times 42 \times 4 \times 1 \times \pi/2}{2} - 3 \{ [16 - 5.33] - [16 + 5.33] \}$$

$$\Rightarrow 527.52 + 3 \times 10.66 = 559.5$$

$$S_1: \hat{n} = \hat{k} \quad x=2$$

$$x_1 = -2 \quad x_2 = 2 \quad y_1 = -2 \quad y_2 = 2$$

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} x^2 dy dx \Rightarrow \int_{-2}^2 dx \int_{-2}^2 dy \Rightarrow 4 \times 4 \times 9 = 144$$

$$\Rightarrow 144 \times 36\pi$$

$$S_2: \hat{n} = -\hat{k}, \quad z=0$$

$$S_3: x^2 + y^2 = 4$$

$$\vec{r} \cdot \hat{n} = 2x\hat{i} + 2y\hat{j}$$

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{(2x)^2 + (2y)^2}} = \frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\vec{r} \cdot \hat{n} = (4x\hat{i} - 2y^2\hat{j} + z\hat{k}) \cdot \left(\frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}} \right)$$

$$\Rightarrow 4x \frac{x}{2} - 2y^3 \frac{1}{2} \Rightarrow 2x^2 - y^3$$

$$\text{R. } \oint \vec{r} \cdot \hat{n} ds \quad x = 2\cos\theta \quad y = 2\sin\theta$$

x varies from 0 to 2 and θ varies from 0 to 2π

$$\int_0^{2\pi} \int_0^2 [2(2\cos\theta)^2 - (2\sin\theta)^3] 2 d\theta dx$$

$$= \int_0^{2\pi} (16\cos^2\theta - 16\sin^3\theta) [x]_0^2 d\theta$$

$$= 48 \int_0^{2\pi} \cos^2\theta - \sin^3\theta d\theta = 48\pi$$

$$\int_S \vec{r} \cdot \hat{n} ds = 36\pi + 48\pi = 84\pi$$

Q3 find $\iint_S \vec{F} \cdot \hat{n} \, ds$ where $\vec{F} = (2x+3z)\hat{i} - (xz+y)\hat{j} + (y^2+2z)\hat{k}$ and S is the surface of the sphere having centre at $(3, -1, 2)$ and radius 3.

$$\sqrt{(x-3)^2 + (y+1)^2 + (z-2)^2} = 3 \quad \left\{ \begin{array}{l} \text{Not} \\ \text{Required} \end{array} \right.$$

$$\nabla \cdot \vec{F} = (2 - 1 + 2)$$

$$\textcircled{2} \frac{2+1+2}{108}$$

$$\iiint_V \nabla \cdot \vec{F} \, dv \Rightarrow 3 \iiint_V dv = 3V = 3 \times \frac{4}{3} \pi r^3 = 4\pi \times 3^3 = 108\pi$$

Q4 Let D be the region bounded by the closed cylinder $x^2 + y^2 = 16$, $z=0$, $z=4$ verify the divergence theorem if $\vec{V} = 3x^2\hat{i} + 6y^2\hat{j} + z\hat{k}$

Ans $\iiint_V \nabla \cdot \vec{V} \, dv = \iint_S \vec{V} \cdot \hat{n} \, ds$

$$\nabla \cdot \vec{V} = 6x + 12y + 1$$

$$\int_{-4}^4 \int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} \int_0^4 (6x + 12y + 1) \, dx \, dy \, dz$$

$$\Rightarrow \int_{-4}^4 \int_0^4 \int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} (6x + 12y + 1) \, dx \, dy \, dz$$

$$\Rightarrow \int_{-4}^4 \int_0^4 \left[3x^2 + 12xy + x \right]_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} \, dy \, dz$$

$$\Rightarrow \int_{-4}^4 \int_0^4 \left[6x^2 + 12yz + x \right]_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} \, dy \, dz$$

$$\Rightarrow \int_{-4}^4 \int_0^4 (6(16-y^2) + 12yz + \sqrt{16-y^2}) \, dy \, dz$$

$$\Rightarrow \int_{-4}^4 \left[96y - 2y^3 + 6yz^2 + \frac{2}{3}(16-y^2)^{3/2} \right]_{y=0}^{y=4} \, dz$$

$$\Rightarrow \frac{16}{4} \int_0^4 dy \times \frac{3 \cdot (16-y^2)}{\sqrt{16-y^2}} + 12x \sqrt{16-y^2} + \sqrt{16-y^2}$$

$$\Rightarrow \frac{16}{4} \int_0^4 dy \int_0^4 (6x + 12y + 1) dy$$

$$\Rightarrow \frac{16}{4} \int_0^4 dy \times \left[6xy + \frac{12y^2}{2} + y \right]_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}}$$

$$\Rightarrow \frac{16}{4} \int_0^4 dy \left[6x\sqrt{16-y^2} + 6(16-y^2) + \sqrt{16-y^2} + 6x\sqrt{16-y^2} - 6(16-y^2) + \sqrt{16-y^2} \right]$$

$$\Rightarrow 4 \times 2 \int_{-4}^4 \sqrt{16-x^2} (6x+1) dx$$

$$\Rightarrow 8 \int_{-4}^4 \sqrt{16-x^2} (6x+1) dx$$

$$\Rightarrow 16 \left[\frac{x}{2} \sqrt{16-x^2} + \frac{16}{2} \sin^{-1}\left(\frac{x}{4}\right) \right]_{-4}^4$$

$$= 16 \times 8 \times \frac{\pi}{2} = 64\pi$$

$$\iint_S \vec{v} \cdot \hat{n} ds = \iint_{S_1} \vec{v} \cdot \hat{n} ds + \iint_{S_2} \vec{v} \cdot \hat{n} ds + \iint_{S_3} \vec{v} \cdot \hat{n} ds$$

On S_1 , $x=0$, $\hat{n} = -\hat{k}$

$$\vec{v} \cdot \hat{n} = 0$$

$$\therefore \iint_{S_1} \vec{v} \cdot \hat{n} ds = 0$$

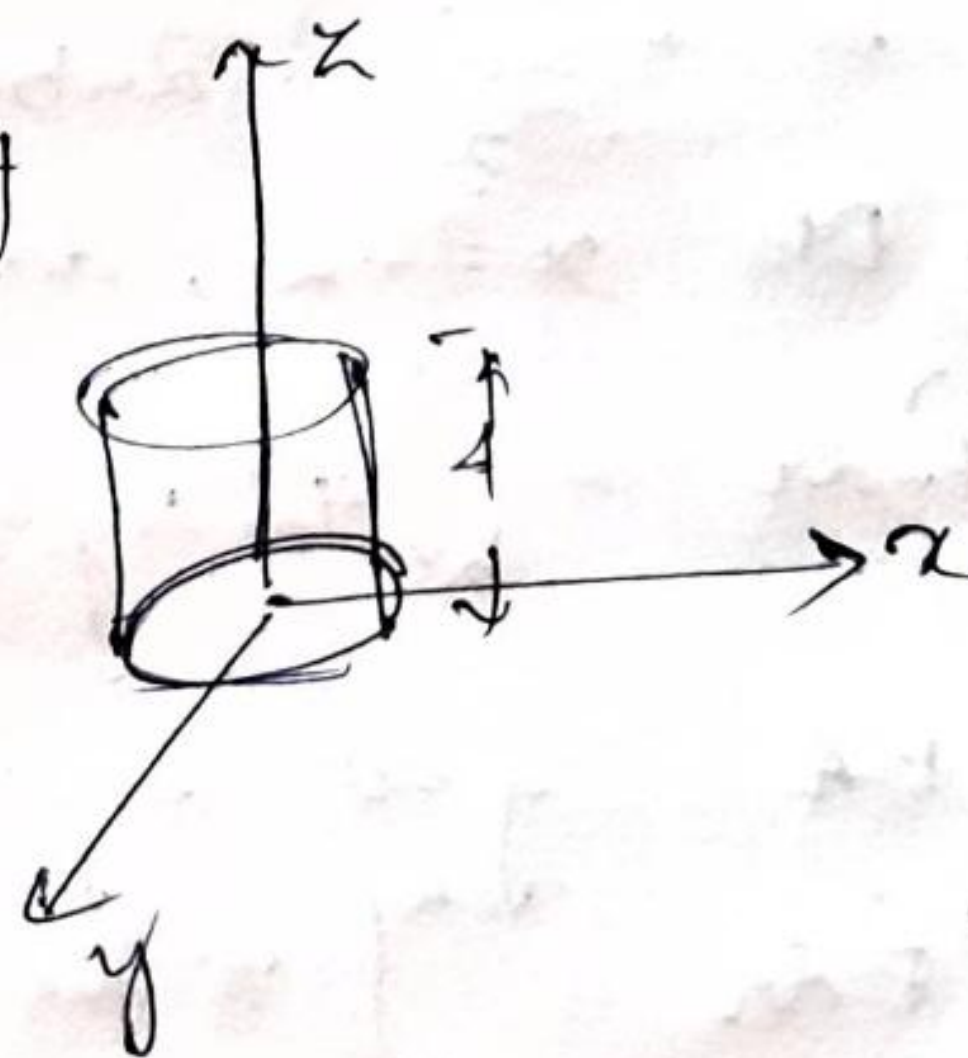
On S_2 , $x=4$, $\hat{n} = \hat{k}$, $\vec{v} \cdot \hat{n} = x = 4$

$$\iint_{S_2} \vec{v} \cdot \hat{n} ds = \iint_{S_2} 4 dxdy$$

$$= 4 \cdot \text{Area} = 4 \times \pi \cdot 4^2 = 64\pi$$

On S_3 , $\hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{x\hat{i} + y\hat{j}}{2}$

$$\vec{v} \cdot \hat{n} = \frac{3x^2}{4} + \frac{6y^2}{4}$$

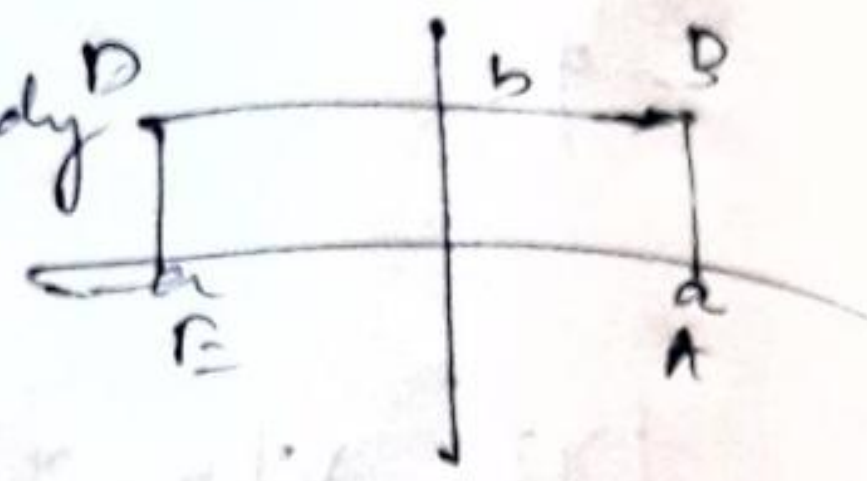


$$Q: \iint_{S_3} \vec{v} \cdot \hat{n} \, ds = \frac{3}{4} \int_{-4}^4 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} x^3 + 2y^3 \, dx \, dy = 0.$$

$$\therefore \iint_S \vec{v} \cdot \hat{n} \, ds = 64\pi$$

9. Verify Stokes' theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken round the rectangle bounded by the lines $x = \pm a, y = 0$.

Ans $\oint_C \vec{F} \cdot d\vec{r} = \oint_C [(x^2 + y^2)\hat{i} - 2xy\hat{j}] \cdot [\hat{i}dx + \hat{j}dy]$



$$= \int_C (x^2 + y^2) dx - 2xy dy$$

Along AB $y = 0, dy = 0, x: a \rightarrow -a$

$$\int_{AB} \vec{F} \cdot d\vec{r} = \int_a^{-a} x^2 dx = -\frac{x^3}{3} \Big|_a^{-a} = -\frac{2a^3}{3}$$

Along BC $x = -a, dx = 0, y: 0 \rightarrow b$

$$\int_{BC} \vec{F} \cdot d\vec{r} = \int_0^b (-2xy) dy = -2x \left[\frac{y^2}{2} \right]_0^b = -ab^2$$

$$= -\frac{2a^3}{3} - ab^2$$

Along CD $y = b, dy = 0, x: -a \rightarrow a$

$$\int_{CD} \vec{F} \cdot d\vec{r} = \int_{-a}^a (x^2 + b^2) dx = \left[\frac{x^3}{3} + b^2 x \right]_{-a}^a = \frac{2a^3}{3} + ab^2$$

Along DA $y = 0, dy = 0, x: a \rightarrow -a$

$$\int_{DA} \vec{F} \cdot d\vec{r} = \int_a^{-a} x^2 dx = -\frac{x^3}{3} \Big|_a^{-a} = -\frac{2a^3}{3}$$

$$\oint_C \vec{F} \cdot d\vec{r} = -\frac{2a^3}{3} - ab^2 + \frac{2a^3}{3} + ab^2 = 0$$

$$= -4ab^2$$

Curl $\vec{F}$:

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix} = -4y\hat{k}$$

$$\int_a^b \int_c^d y \, dx \, dy = \int_0^b \left[-\frac{1}{2}xy^2 \right]_c^d dy$$

$$-\frac{1}{2} \int_0^b y^2 dy = -\frac{1}{6}ab^2$$

$$\int \vec{F} \cdot \hat{n} \, ds$$

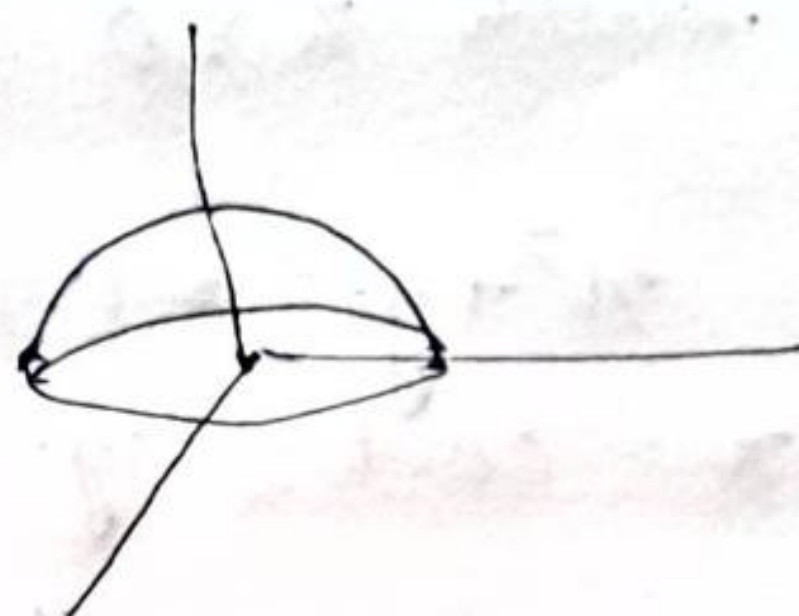
then for the vector field

$\vec{F} = yz^2 \hat{i} - yz^2 \hat{j} - y^2 z \hat{k}$ Over the upper half
 $(x^2 + y^2 + z^2 = 1)$ bounded by its projection on

$$x^2 + y^2 = 1 \quad z=0$$

$$y = \sin t$$

$$t: 0 \rightarrow 2\pi$$



$$\left[(2xy) \hat{i} - yz^2 \hat{j} - y^2 z \hat{k} \right] \cdot \left[\hat{i} \, dx + \hat{j} \, dy + \hat{k} \, dz \right]$$

$$x^2 + y^2 + z^2 = 1$$

$$y \, dx - yz^2 \, dy - y^2 z \, dz = \int (2xy) \, dx$$

$$\int_0^{2\pi} (2 \cos t - \sin t) - \sin t \, dt = \int_0^{2\pi} -\sin t \, dt + \sin^2 t \, dt$$

$$= -\sin t + \frac{1 - \cos 2t}{2} \, dt = \left[\frac{\cos 2t}{2} + \frac{1}{2} - \sin \frac{2t}{2} \right]_0^{2\pi}$$

$$= \frac{1}{2} + \frac{2\pi}{2} - 0 - \frac{1}{2} + 0 = \pi$$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & -yz^2 & -y^2 z \end{vmatrix} = \hat{i} [-2yz + 2yz] - \hat{j} [0 - 0] + \hat{k} [0 + 1]$$

$$\text{curl } \vec{F} \cdot \hat{n} = \hat{k} \cdot \hat{n}$$

$$\int_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_R \hat{k} \cdot \hat{n} \frac{dxdy}{(\hat{n} \cdot \hat{k})} = \int_R dxdy$$

$$\begin{aligned}
 \text{Q1} \quad \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds &= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_0^1 dy \, dx = \int_0^1 2\sqrt{1-x^2} \, dx \\
 &= 2 \int_0^1 \sqrt{1-x^2} \, dx = 2 \left[x \frac{\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1} x \right]_0^1 = \frac{1}{2} (\sin^{-1} 1) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

Q2 Evaluate the surface integral $\iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$ by transforming it into a line integral $\oint_C$ using the part of the surface of the paraboloid $z = 1 - x^2 - y^2$ for which $z \geq 0$ and $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$.

ans $x^2 + y^2 = 1, z = 0$
 $x = \cos t, y = \sin t, z = 0 \quad 0 \leq t \leq 2\pi$

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \oint_C \vec{F} \cdot d\vec{r} = \oint_C (y\hat{i} + z\hat{j} + x\hat{k}) (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \oint_C y \, dx + z \, dy + x \, dz = \oint_C y \, dx$$

$$= \int_0^{2\pi} \sin t \cdot \sin t \, dt = - \int_0^{2\pi} \sin^2 t \, dt = - \int_0^{2\pi} \frac{1 - \cos 2t}{2} \, dt$$

$$= - \left[\frac{t}{2} - \frac{\sin 2t}{4} \right]_0^{2\pi} = - \frac{2\pi}{2} = -\pi$$